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The current progress, trend, and challenges of immunoglobulin for therapeutics

1. Introduction

Immunoglobulin (Ig), known as an antibody, is one kind of large protein with a Y-shaped structure, which stems from plasma cell and aims to help the immune system to eliminate outcome pathogens such as bacteria, infected cells, viruses. These pathogens with unique molecules are named antigen, and they are often recognized by the fragment antigen-binding variable region of immunoglobulin.[1] As shown in the left image of Figure 1, we can find the existence of paratope works as a lock, which means only one kind of antibody can combine with one specific antigen via the interaction of non-covalent bonds such as hydrogen bond, Van Der Waals force and even hydrophobic interaction.

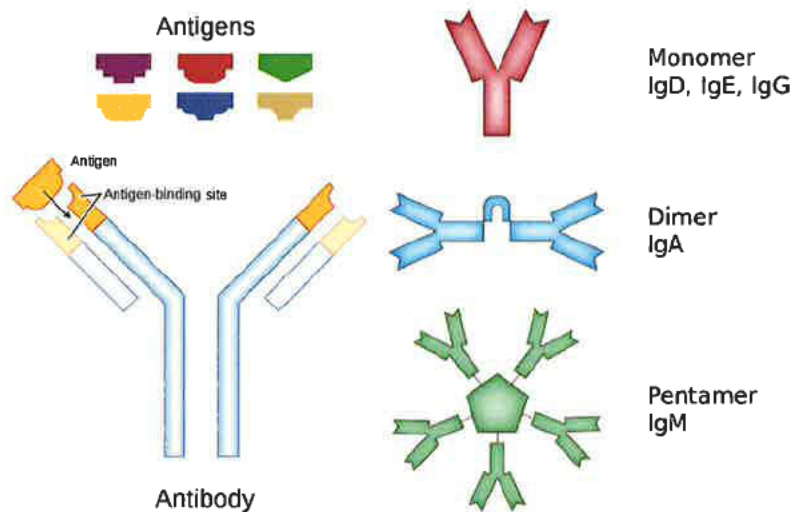


Figure 1 The scheme of the different structure of 5 antibodies [1]

According to the various function, these immunoglobins (Ig) can be divided into 5 different subclasses, and they are marked as IgA, IgD, IgE, IgG, and IgM. The right part of Figure 1 shows three different structures of 5 antibodies. Each one kind of antibody have different functions, and the type of B cell antitypes will change with the development of itself. Firstly, immature B cells will change into the type of IgM, which is a pentamer with really high avidity. It is used to eliminate pathogens when the primary IgG is insufficient at the early stage of invasion. As the B cell starts to be mature, it will generate two types, IgM, and IgD, which imply that at this time B cell has had an ability to fight with antigen. There is an existing mechanism that can generate IgE and IgA.[1]

Basically, due to the unique function of the antibody, many researchers have tried to develop a related drug to treat cancer, which is called antibody-based immunotherapy. During the process of treatment, these well-developed antibodies can break the signal transferring process of different growth factor receptors by tracing and aiming to the surface antigens, which are identified by cells.[2] In this report, it will introduce the recent current progress, trend, and challenges of antibodies drugs that mostly will be

used to treat cancer.

2. Current progress and trend

In the past 20 years, as a significant component of standard cancer treatment, antibodies have been proved to own one specific ability to identify those surface antigens, which will be expressed by tumor cells. And, itself acts as one kind of popular and passive therapy method. For example, two types of factor receptors, epidermal growth factor receptor (EGFR) and human epidermal growth factor receptor 2 (HER2) are common receptors that often used to treat non-leukemia. There is one type of antibody that targets the lineage markers of hematopoietic cells, such as CD20, presents highly efficient treatment for hematological malignancies. The mechanism between the process is that these antibodies can combine with the membrane of receptors, which results in the decrease of the signaling process of tumor cell and finally eliminate tumor cells. This is because the signaling process plays a vital role in promoting the cycle and function of tumor cells, and the Fc-mediated innate immunological effector mechanism like antibody-dependent cell-mediated cytotoxicity (ADCC) is affected and even promoted by that signaling process as well. Generally, if we can obtain the basic mechanism of antibody drugs, the novel therapy of using antibody drugs will help doctors open a brand-new way to treat patients with cancer.

As we all know, antibodies are made up of two identical light chains and heavy chains with two distinct functional units, which are the fragment of antigen-binding (Fab) and the constant fragment (Fc). Each variable region of that light and the heavy chain will combine together and form the so-called antigen binding site. Thus, one antibody has two special antigen-binding sites.

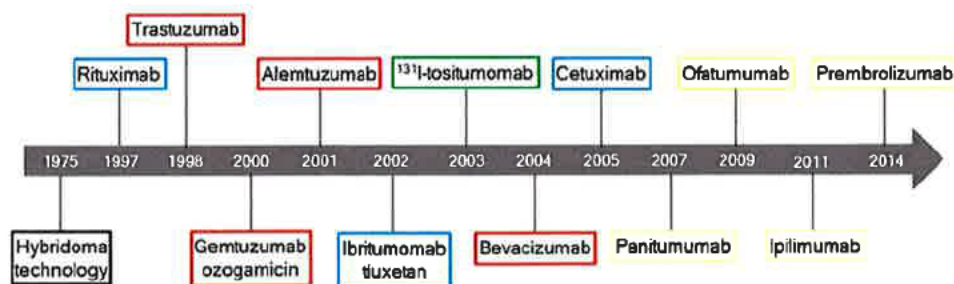


Figure 2 The timeline of immunoglobulin development for cancer treatment [2]

As shown in Figure 2, we can find the history that humans began to use antibody as a biological tool that starts in 1975. At that time, it was monoclonal antibodies that became the most useful medicine when doing a diagnosis on the patient. [3] However, the clinical results of an early trial made human disappointed. Although the first monoclonal antibody has been proved to be successful on the mouse, it always causes some serious immune response on the human due to the immunogenicity, which hampers the further application on the clinic. In 1997, as the first antibody that was used to treat cancer, rituximab, one kind of mouse chimeric monoclonal antibody against the B cell

lineage labeled CD20, is synthesized and get approved by the FDA for the treatment of B cell non-Hodgkin's lymphoma. After that, many researchers had made an effort to study and develop new antibody drugs aiming to relieve the painlessness of cancer or eliminate the tumor cells.[4] Up to 2014, antibodies are also divided into several types due to their different source. For instance, it exists the type of chimeric antibodies such as Rituximab, Ibritumomab tiuxetan, and Cetuximab. And humanized antibodies, like Trastuzumab, Gemtuzumab ozogamicin, Alemtuzumab, and Bevacizumab, are developed at the beginning of 2000. So far, there is another kind of antibody called human antibody appearing in the latest 10 years. The famous ones among those antibodies are Panitumumab, Ipilimumab, Ofatumumab, Pembrolizumab.[2] These antibody drugs are fully utilized to fight against cancers, and the immune system is activated to the most active status in order to decrease the tumor cells.

Actually, the antibody is classified into two different types because of their target, thus resulting in two various therapies. One is a tumor-targeted antibody, the other is an immunomodulating antibody.

The tumor-targeted antibody has been used in the treatment for cancer for over 30 years, and the clinic results show this method has relatively high efficacy and act like the most successful strategy for the hematological malignancies and solid tumors. The nature of targets of tumor cells are growth factor receptors and antigens, and then they will be further divided into the epidermal growth factor receptor (EGFR), human epidermal growth factor receptor 2 (HER2), vascular endothelial growth factor (VEGF), cluster of differentiation (CD) with subclass like CD20, CD52, CD19, and CD30.[5] Basically, all functions of the antibody rely on the interaction of these mentioned receptors or antigens.

The real mechanism that tumor-targeted antibodies work on the tumor cells have been proposed and explained in the past 30 years. It can be concluded by 4 functions. The first one is the function of blocking the ligand binding and signaling. It can be easily explained by the overexpressed growth factor receptors. If something can block the connection between ligands and receptors or send signals generated by those receptors, the monoclonal antibody will let the tumor cell grow back to the normal style or let them die automatically.[6] The second one is called Antibody-Dependent Cellular Cytotoxicity and Phagocytosis. Briefly, if antibodies can combine with their cognate antigen, which has to be expressed on the same tumor cell, their constant fragments will connect with the related constant fragment receptor sites, which are grown on monocytes and granulocytes, to activate the function of antibody-mediated cellular cytotoxicity (ADCC), and finally contributing to the self-destruction of tumor cells.[7] The next mechanism is the function of Complement-Dependent Cytotoxicity. In nature, it is just another expression of the second mechanism. The ability of specific antibodies that induce cytotoxicity actually depends on the concentration and direction of antigen in the membrane. However, it is generally accepted that the third one has a fewer influence on explaining the nature of tumor-based antibodies when compared with the second

one.[8] The final mechanism is the Induction of T-Cell Immunity Through Cross-Presentation. There is an increasing amount of evidence, both from the animal model and clinical results, showing that the immune responses of those tumor cells can be enhanced by the appearance of tumor-targeted antibodies, including cytotoxic T lymphocytes.[9] Unlike the previous ways of blocking the signaling of tumor cells, this new way provides a new possibility that antibody has a direct ability to stimulate the immune response of tumor cells, which increases the efficiency of antigen cross-presentation.

Apart from the tumor-targeted antibody mentioned in the previous paragraph, many scholars have found a new working function of the antibody, which aims to enhance the specific antitumor immunity of the whole immune system.[10] Although some of this kind of antibody has been proved to have a good treatment effect, it also exists unknown side effects and needs to be done more researches on animal models or clinical trials.

3. Challenges

Although the antibody for cancer therapy shows an exciting treatment effect for tumor cells, it still needs to do more quantified trials to define the best activity if a human wants to fully develop it as the most standard and newest therapy for cancers. Besides, due to the unique properties of the antibody, it should be extended to more types of cancers rather than those cancers that have been studied many times. In addition, the exploration of the antibody-drug is far less than the level that can be fully promoted to the patient. More properties, including the round number of replication and side effects of innate immune cell suppression, are all need to be confirmed in order to fully used in the clinic.

There are many unknown effects or mechanisms waiting to be discovered and proved by scientists. Admittedly, antibody treatment for cancer brings about a new strategy for patients and giving their new hopes for living. With the rapid development of modern medicine and related subjects, this field becomes more and more fascinating.

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